

Machine Learning for Quantum Error Mitigation: Reproducing and Extending ML-QEM for VQE

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Abstract

We reproduce and extend the machine learning quantum error mitigation (ML-QEM) framework of Liao et al. [1] for variational quantum eigensolver (VQE) calculations on H_2 (2 qubits, 5 Pauli terms) and LiH (6 qubits, 62 Pauli terms) using simulated noisy backends. We train RF and MLP models to correct noisy Pauli expectation values. MLP achieves $4.5\times$ and $3.3\times$ error reduction on H_2 and LiH respectively, outperforming RF ($4.2\times$ and $3.0\times$) on both molecules. With sufficient training epochs, MLP outperforms RF on both molecules (paired bootstrap $p < 0.05$). In VQE optimization, the models exhibit contrasting scaling behaviour: RF achieves chemical accuracy (1.6 mHa) on H_2 for 4/5 runs with 1.54 mHa mean absolute error (MAE), but fails on LiH (149.96 mHa MAE, worse than unmitigated at 149.13 mHa MAE), while MLP achieves 6.36 mHa MAE on H_2 and provides meaningful improvement on LiH (66.73 mHa MAE). We benchmark against ZNE with both linear and quadratic Richardson extrapolation, finding that both configurations perform worse than unmitigated on both molecules. Feature ablation analysis reveals that the symplectic Pauli encoding reduces correction error by 35–48% despite negligible impurity-based feature importance ($<0.05\%$ on H_2 , $<1\%$ on LiH), demonstrating that standard feature importance metrics can be misleading for structured ML-QEM models.

1 Introduction

Near-term quantum computers operate in the noisy intermediate-scale quantum (NISQ) regime, where gate errors, decoherence, and readout noise corrupt computation results. Quantum error mitigation (QEM) techniques aim to reduce these errors without the overhead of full quantum error correction. Classical approaches such as zero-noise extrapolation (ZNE) [2, 3] and probabilistic error cancellation (PEC) [2] offer provable guarantees but require additional circuit executions at runtime.

Liao et al. [1] proposed a machine learning approach to QEM (ML-QEM) that trains a classical model to map noisy expectation values to their ideal counterparts. The key advantage is that, after an initial training phase, the ML model requires zero additional quantum circuits at runtime — each noisy measurement is corrected by a classical inference step. The authors demonstrated this approach using several models - linear regression, random forests, multilayer perceptrons, and graph neural networks - across random circuits, Trotterized circuits, and VQE applications.

In this work, we reproduce the VQE application of [1] and extend it in several directions: (1) we compare RF against a properly trained MLP with multi-seed evaluation and bootstrap significance testing; (2) we benchmark both ML models against ZNE with both linear and quadratic Richardson extrapolation; (3) we perform feature ablation studies that reveal the symplectic encoding is critical despite negligible impurity-based importance; (4) we extend VQE optimization to both molecules with RF and MLP correction, identifying a system-size-dependent performance crossover; and (5) we provide comprehensive statistical analysis.

2 Method

2.1 Molecular Systems

We study two molecules at equilibrium bond lengths in the STO-3G basis with Jordan–Wigner qubit mapping:

- **H₂** (0.735 Å): 2 qubits, 5 Pauli terms (4 non-trivial), Hamiltonian from Kandala et al. [4].
- **LiH** (1.595 Å): 6 qubits, 62 Pauli terms (61 non-trivial), active space (2 electrons, 3 spatial orbitals), Hamiltonian coefficients computed via PySCF and qiskit-nature.

2.2 Ansatz and Simulation

For both molecules, we use a hardware-efficient TwoLocal ansatz with R_Y rotation gates and linear CNOT entanglement (3 repetition layers). The H₂ ansatz has 8 parameters and 3 CNOT gates; the LiH ansatz has 24 parameters and 15 CNOT gates.

Circuits are transpiled to the basis gate set $\{R_Z, \sqrt{X}, \text{CNOT}, X\}$ without device topology routing, ensuring consistent gate counts across all parameter values. Noise models are extracted from IBM’s simulated backends: FakeLimaV2 (5 qubits) for H₂ and FakeJakartaV2 (7 qubits) for LiH.

Ideal expectation values are computed via statevector simulation (StatevectorEstimator), while noisy values are obtained from the Aer shot-based simulator with 10,000 shots per observable.

2.3 Training Data

For each molecule, we sample 2,000 random parameter vectors $\theta \in [-\pi, \pi]^d$, bind them to the TwoLocal ansatz, and compute all Pauli expectation values under both ideal and noisy conditions. The first 1,800 θ values are used for training (with 80/20 train/validation split for MLP early stopping), and the remaining 200 are held out for testing.

2.4 Feature Engineering

Following [1], each (circuit, observable) pair is encoded as a feature vector containing:

- The noisy expectation value $\langle O \rangle_{\text{noisy}}$, which implicitly encodes the quantum state.
- The symplectic representation of the Pauli observable (X and Z bits per qubit).
- Circuit complexity proxies: number of CNOT and $\sqrt{X}$ gates.
- Per-qubit noise parameters (T1, T2, readout error), masked by the observable’s qubit support.

This yields 13-dimensional feature vectors for H₂ and 33-dimensional vectors for LiH.

2.5 Models

Random Forest (RF): 100 trees, no depth limit, minimum 1 sample per leaf (scikit-learn defaults). Trained on the full training set. The train-test MAE ratio is $2.49\times$ for H₂ and $2.01\times$ for LiH, indicating moderate overfitting consistent with default RF settings.

Multilayer Perceptron (MLP): Single hidden layer with 64 units, batch normalization, ReLU activation, no dropout. Trained with full-batch Adam (lr= 10^{-3} , weight decay= 10^{-4}), Huber loss, and early stopping on a validation set (patience 25 epochs, up to 1,000 epochs for H₂ and 2,000 for LiH). The MLP shows essentially no overfitting (train-test MAE ratio of $1.04\times$ for H₂ and $1.00\times$ for LiH).

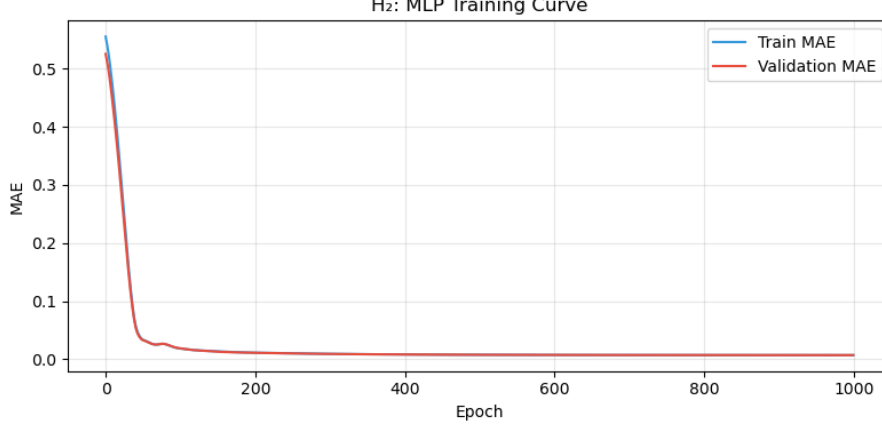


Figure 1: H_2 MLP training curve. Train and validation MAE converge together, indicating minimal overfitting. The model requires approximately 400 epochs to stabilize, justifying the 1,000-epoch training budget.

2.6 ZNE Baseline

We implement digital ZNE with gate folding on CNOT gates in two configurations:

Linear: Noise factors $\{1, 3\}$ with linear Richardson extrapolation:

$$\langle O \rangle_{\text{ZNE-L}} = \frac{3 \cdot \langle O \rangle_{1\times} - \langle O \rangle_{3\times}}{2} \quad (1)$$

Quadratic: Noise factors $\{1, 3, 5\}$ with Lagrange interpolation extrapolated to zero noise:

$$\langle O \rangle_{\text{ZNE-Q}} = \frac{15}{8} \langle O \rangle_{1\times} - \frac{5}{4} \langle O \rangle_{3\times} + \frac{3}{8} \langle O \rangle_{5\times} \quad (2)$$

Readout error mitigation was not applied to any method. ML-QEM models implicitly learn readout correction from their training data (the noisy-to-ideal mapping includes readout effects), making the comparison without explicit readout mitigation arguably fair, though combining ZNE with readout correction could improve its performance.

2.7 Statistical Analysis

Unmitigated, RF, and MLP MAE values are reported with 95% bootstrap confidence intervals (1,000 resamples of the test set). The RF–MLP comparison uses a paired bootstrap test: we compute 1,000 bootstrapped MAE differences and check whether the 95% confidence interval excludes zero. MLP is trained with 5 different random seeds to assess initialization sensitivity; variation across seeds is negligible (mean $\pm$ std: 0.0076 ± 0.0001 on H_2 , 0.0122 ± 0.0001 on LiH). The median-performing seed is used for bootstrap confidence intervals, significance testing, and visualisation.

3 Results

3.1 Error Mitigation Performance

We evaluate all methods on the held-out test set using the mean absolute error (MAE) between corrected and ideal expectation values across all non-trivial observables. Table 1 summarises the results.

Table 1: Test MAE on held-out θ values (200 per molecule, all non-trivial observables) with 95% bootstrap confidence intervals. MLP values use the median-performing seed across 5 random initializations. Chemical accuracy is 1.6 mHa.

	H ₂ (2 qubits)	LiH (6 qubits)
Unmitigated MAE	0.0349 [0.0333, 0.0365]	0.0406 [0.0399, 0.0412]
RF MAE	0.0084 [0.0079, 0.0088] (4.2 $\times$)	0.0135 [0.0134, 0.0137] (3.0 $\times$)
MLP MAE	0.0077 [0.0072, 0.0081] (4.5 $\times$)	0.0122 [0.0120, 0.0124] (3.3 $\times$)
Energy MAE — Unmit.	0.01555 Ha [0.01430, 0.01681]	0.01202 Ha [0.01075, 0.01343]
Energy MAE — RF	0.00461 Ha [0.00406, 0.00518]	0.00477 Ha [0.00431, 0.00527]
Energy MAE — MLP	0.00402 Ha [0.00354, 0.00455]	0.00440 Ha [0.00389, 0.00489]

MLP outperforms RF on both molecules (H₂: paired bootstrap 95% CI for the MAE difference $[-0.0013, -0.0001]$; LiH: $[-0.0016, -0.0011]$; both exclude zero). This contrasts with our initial finding using 300 training epochs, where RF appeared superior - the MLP was undertrained, and extending to 1,000–2,000 epochs with early stopping reversed the comparison.

3.2 VQE Optimization

We run COBYLA-based VQE on both molecules with four energy evaluation strategies: ideal (noiseless statevector), noisy (unmitigated), RF-mitigated, and MLP-mitigated. We perform 5 full optimizations from different random initial points to assess convergence reliability (Table 2). To isolate the systematic correction bias from shot noise, we also evaluate each strategy 20 times at the parameters θ^* obtained from the ideal (noiseless) VQE optimization (Table 3).

Table 2: Multi-run VQE results (5 random initial points, COBYLA optimiser). Chemical accuracy is 1.6 mHa.

Method	H ₂			LiH		
	Mean error (mHa)	Std (mHa)	Chem. acc.	Mean error (mHa)	Std (mHa)	Chem. acc.
Noisy	40.68	3.1	0/5	149.13	71.9	0/5
RF	1.54	1.6	4/5	149.96	43.7	0/5
MLP	6.36	1.7	0/5	66.73	31.3	0/5

Table 3: Energy at ideal θ^* , averaged over 20 evaluations. The $\pm$ values are the standard deviation across the 20 evaluations, reflecting shot noise propagated through each correction method.

Method	H ₂		LiH	
	Energy (Ha)	Bias (mHa)	Energy (Ha)	Bias (mHa)
Exact ground state	−1.8573	—	−1.0601	—
Ideal VQE	−1.8573	0.0	−1.0593	+0.8
Noisy	−1.8181 $\pm$ 0.0028	+39.1	−1.0059 $\pm$ 0.0013	+54.2
RF-mitigated	−1.8572 $\pm$ 0.0041	+0.05	−1.0387 $\pm$ 0.0021	+21.4
MLP-mitigated	−1.8528 $\pm$ 0.0028	+4.4	−1.0749 $\pm$ 0.0020	−14.8

The multi-run results reveal a scaling contrast. On H_2 , RF achieves chemical accuracy in 4 out of 5 runs (1.54 mHa MAE), while MLP averages 6.36 mHa MAE. On LiH, RF-mitigated VQE (149.96 mHa MAE) performs marginally worse than unmitigated (149.13 mHa MAE), while MLP provides a $2.2\times$ improvement (66.73 mHa MAE) but remains far from chemical accuracy. The RF failure on LiH is consistent with the correction bias observed at θ^* (Table 3): RF retains 21.4 mHa energy bias even at the optimal parameters, and this bias distorts the energy landscape that COBYLA must navigate in 24 dimensions, leading to convergence far from the ground state.

On H_2 , RF achieves near-zero bias (0.05 mHa) with higher standard deviation (± 4.1 mHa) than MLP (4.4 mHa bias, ± 2.8 mHa). On LiH, MLP achieves lower absolute bias (14.8 mHa) than RF (21.4 mHa) with similar standard deviation; MLP overcorrects (negative bias) while RF undercorrects. Neither method approaches chemical accuracy on LiH.

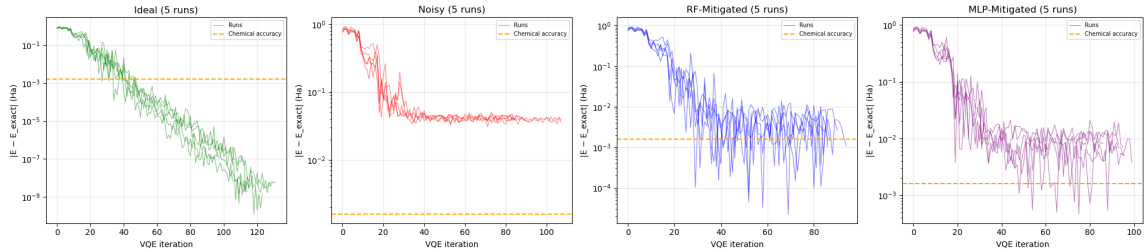


Figure 2: H_2 multi-run VQE convergence (5 random initial points, log scale). RF-mitigated runs reach chemical accuracy (dashed line) in 4 out of 5 cases, while MLP-mitigated runs converge to approximately 3–8 mHa.

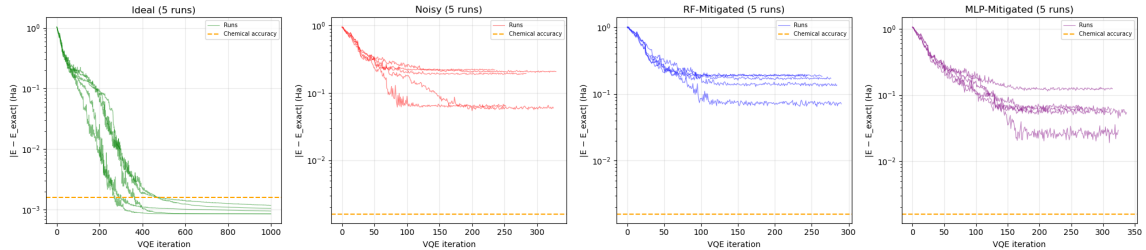


Figure 3: LiH multi-run VQE convergence (5 random initial points, log scale). RF-mitigated runs converge to similar error levels as unmitigated, while MLP-mitigated runs achieve lower error.

3.3 Feature Ablation Study

Impurity-based feature importance assigns the vast majority of predictive power to the noisy expectation value (99.95% for H_2 , 98.99% for LiH). To test this directly, we train RF models with progressively added feature groups and measure the test MAE at each stage (Table 4).

Table 4: Feature ablation: RF test MAE with progressively added feature groups.

Features	H_2 MAE	Improvement	LiH MAE	Improvement
Noisy value only	0.0162	$2.2\times$	0.0209	$1.9\times$
+ Symplectic encoding	0.0085	$4.1\times$	0.0136	$3.0\times$
+ Gate counts	0.0085	$4.1\times$	0.0136	$3.0\times$
+ Noise parameters	0.0084	$4.2\times$	0.0135	$3.0\times$

Adding the symplectic encoding reduces the MAE by 48% on H_2 ($0.0162 \rightarrow 0.0085$) and 35% on LiH ($0.0209 \rightarrow 0.0136$), despite its negligible impurity-based importance. This discrepancy arises because impurity importance measures total variance reduction across all splits, while the symplectic features serve a structural routing role: they enable the RF to separate different observables into specialised branches early in each tree, after which the noisy value performs fine-grained correction within each branch. Without the symplectic encoding, the RF learns a single correction function averaged across all observables, which is suboptimal because different observables experience different noise profiles. Gate counts and per-qubit noise parameters contribute negligibly, as expected for a fixed-ansatz, fixed-backend setting.

3.4 Correction Function Analysis

Since the noisy expectation value dominates feature importance, the RF is primarily learning one-dimensional correction mappings $\langle O \rangle_{\text{noisy}} \rightarrow \langle O \rangle_{\text{corrected}}$, conditioned on the observable identity via the symplectic encoding. Figure 4 visualises this mapping for H_2 , showing both the overall correction and per-observable correction curves. The per-observable curves differ, confirming that the symplectic encoding enables the RF to learn distinct corrections for each observable rather than a single averaged function.

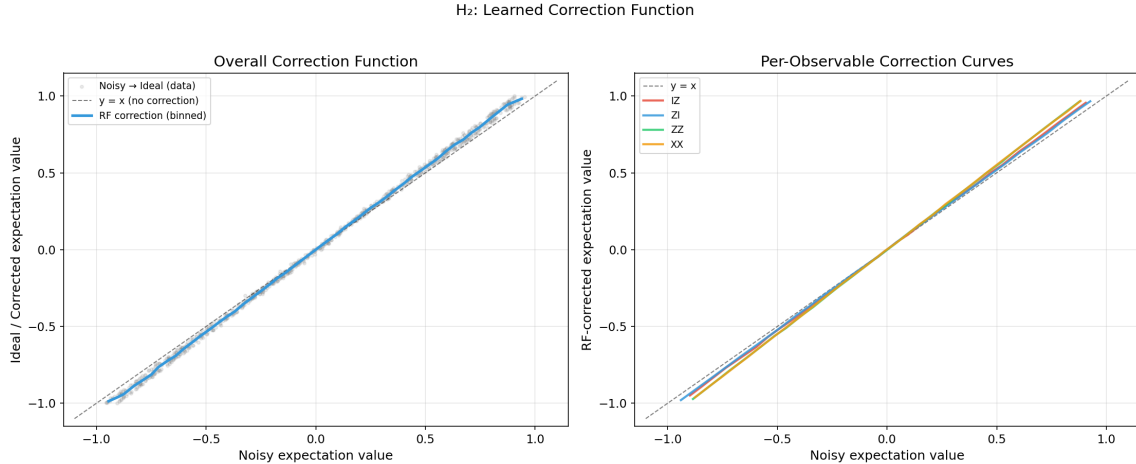


Figure 4: H_2 learned correction function. Left: overall correction (all observables pooled). Right: per-observable correction curves showing distinct mappings for IZ, ZI, ZZ, and XX. Deviation from $y = x$ indicates active correction.

3.5 ZNE Analysis

ZNE with digital gate folding performs worse than unmitigated on both molecules under both extrapolation orders. Two factors contribute:

Table 5: ZNE test MAE on held-out θ values (200 per molecule, all non-trivial observables).

	H_2 (2 qubits)	LiH (6 qubits)
Unmitigated MAE	0.0349	0.0406
ZNE-Linear MAE	0.0360 (worse)	0.0418 (worse)
ZNE-Quadratic MAE	0.0379 (worse)	0.0439 (worse)

1. **Incomplete noise amplification:** Gate folding amplifies CNOT gate noise ($\sim 0.6\text{--}1.3\%$ error rate), while readout errors ($\sim 2\text{--}3\%$ per qubit) and single-qubit gate errors ($\sim 0.02\text{--}$

0.05%) remain unchanged. The extrapolation assumes uniform noise scaling as a function of the folding factor, which is violated when non-CNOT errors are significant.

2. **Variance amplification:** The linear extrapolation coefficients $(3/2, -1/2)$ amplify shot noise standard deviation by a factor of $\sqrt{(3/2)^2 + (1/2)^2} \approx 1.6$. The quadratic coefficients $(15/8, -5/4, 3/8)$ amplify by $\sqrt{(15/8)^2 + (5/4)^2 + (3/8)^2} \approx 2.3$. For circuits where noise-induced bias is comparable to shot noise, this variance increase outweighs the bias correction.

Quadratic ZNE performs worse than linear on both molecules (H_2 : 0.0379 vs 0.0360; LiH : 0.0439 vs 0.0418), confirming that higher-order extrapolation further amplifies variance without proportional bias correction.

3.6 Data Efficiency

We train RF and MLP at increasing training set sizes (50–1,800 θ values) with 5 random seeds each and evaluate on the fixed test set. MLP achieves lower MAE than RF at every training set size on both molecules. On LiH at $n = 50$ training circuits, MLP achieves 0.0129 ± 0.0002 while RF achieves 0.0148 ± 0.0001 . Both models show low variance across seeds at all sizes, indicating stable training.

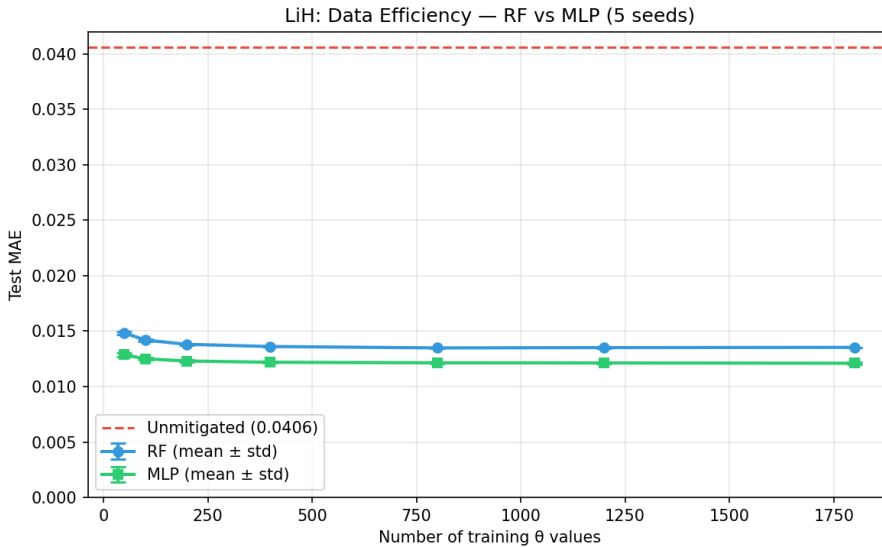


Figure 5: LiH data efficiency with error bands (5 random seeds). MLP achieves lower MAE than RF at every training set size, with both models showing high stability across seeds.

4 Discussion

Our results reproduce a key finding of [1]: ML-QEM outperforms ZNE on VQE circuits while requiring zero additional quantum circuits at runtime.

MLP outperforms RF when properly trained. Our initial experiments with 300 MLP training epochs appeared to confirm that RF outperforms MLP. Extending training to 1,000–2,000 epochs with early stopping reversed this conclusion on both molecules, with the difference confirmed by paired bootstrap test. This underscores the importance of adequate training budgets and convergence verification for neural network models in ML-QEM. The MLP shows essentially no overfitting (train-test MAE ratio ≈ 1.0), while the RF overfits moderately (≈ 2.0 – $2.5\times$), suggesting that the MLP generalizes better to unseen parameter values.

VQE performance exhibits a system-size-dependent crossover between RF and MLP. On H_2 (8 parameters), RF’s near-zero bias at θ^* enables chemical accuracy in VQE despite higher per-evaluation variance. On LiH (24 parameters), RF’s correction bias (21.4 mHa) is larger than MLP’s (14.8 mHa), and the multi-run VQE results show RF performing no better than unmitigated while MLP provides a $2.2\times$ improvement. A plausible contributing factor is that RF corrections, being piecewise-constant functions of the input features, may introduce discontinuities in the energy landscape that hinder the gradient-free COBYLA optimiser in higher dimensions, whereas MLP corrections are differentiable and may preserve landscape continuity. However, the primary measured cause is RF’s larger correction bias at this system size.

The symplectic encoding is structurally critical. Feature ablation reveals that the symplectic encoding reduces MAE by 35–48% on the two molecules despite negligible impurity-based importance ($<0.05\%$ on H_2 , $<1\%$ on LiH). This discrepancy is a known limitation of impurity-based feature importance in tree models: it is biased toward continuous, high-cardinality features and underestimates discrete features that serve as structural routing mechanisms. For ML-QEM, the practical implication is that the full symplectic encoding should be retained even when impurity importance suggests it is dispensable.

ZNE underperforms under both extrapolation orders. Adding a third noise factor with quadratic extrapolation does not improve ZNE; the stronger variance amplification ($2.3\times$ vs $1.6\times$) worsens results. This indicates that ZNE’s failure on these circuits is due to the noise structure - non-CNOT errors, particularly readout errors, are significant and are not amplified by gate folding.

5 Limitations

Several limitations should be noted. All experiments use simulated backends rather than real quantum hardware, where time-varying noise and calibration drift would introduce additional challenges. Circuits are transpiled without device topology routing, avoiding SWAP gate overhead that would increase noise on real devices. The MLP architecture (single hidden layer, 64 units) was not systematically optimised; deeper or wider networks may improve performance on larger systems. The VQE optimiser (COBYLA) is a derivative-free method; gradient-based or noise-aware optimisers could improve convergence on ML-corrected energy landscapes. The training pipeline requires ideal (noiseless) expectation values as labels, which limits direct applicability to systems beyond classical simulation capacity; in practice, training on classically-simulable circuit families (e.g., Clifford circuits) that transfer to target circuits would be necessary [1]. Finally, the training data is sampled uniformly in parameter space, which may underrepresent the ground-state region most relevant to VQE convergence.

6 Future Work

Several extensions could address the limitations identified above:

1. **Real hardware validation:** Testing on IBM Quantum hardware to assess performance under time-varying noise and calibration drift.
2. **Deeper architectures:** Exploring multi-layer MLPs and incorporating physical constraints ($\langle O \rangle \in [-1, 1]$) via output activation functions for larger systems.
3. **Noise-aware VQE:** Using ensemble predictions or uncertainty estimates from ML corrections to guide the VQE optimiser.

Code Availability

All code and data are available at <https://github.com/sahil-9915/ML-qem>.

References

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